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Winter et al.

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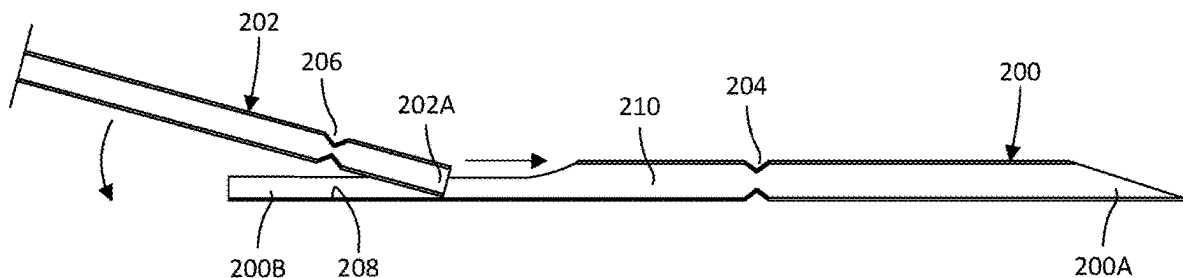
(57) **ABSTRACT**

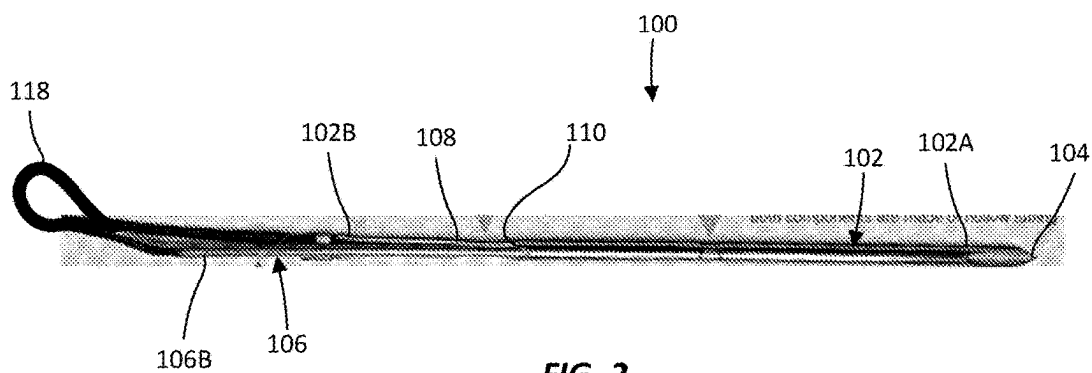
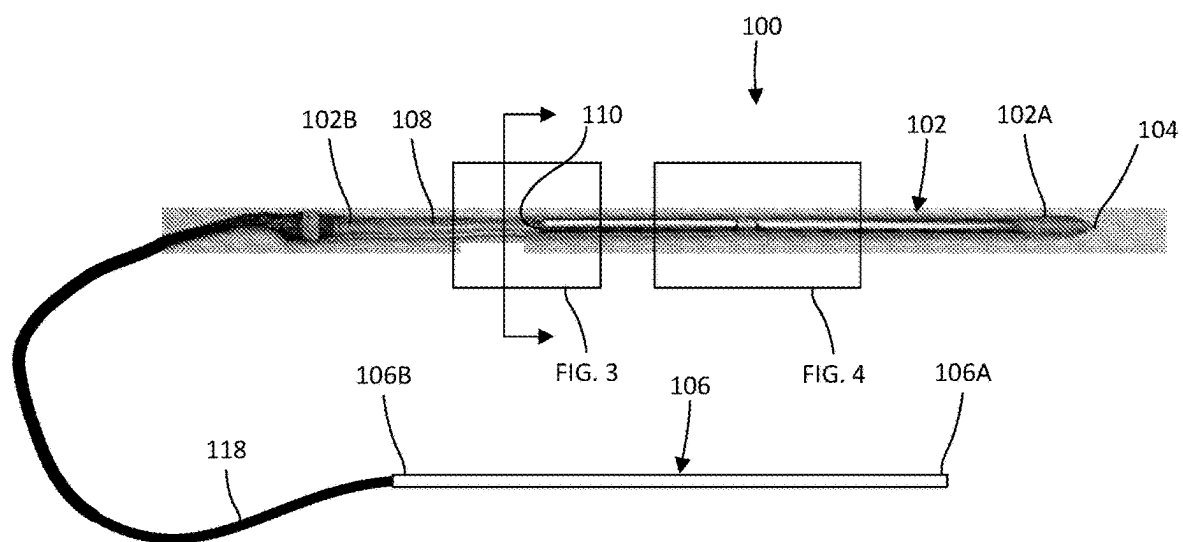
An interlocking two-part sewing needle includes a first needle portion having a needle tip and a second needle portion configured to be inserted into the first needle portion for removably connecting the first needle portion and the second needle portion together. A needle receiver is formed in the first needle portion and is configured to receive the second needle portion when the second needle portion is inserted into the first needle portion. The needle receiver includes one or more non-resistance sections. Each non-resistance section is configured to exert only a friction force on the second needle portion when the second needle portion is inserted into the needle receiver. Additionally, the needle receiver includes one or more resistance sections. Each resistance section is configured to exert a compressive force on the second needle portion when the second needle portion is inserted into the needle receiver.

(22) Filed: **Mar. 14, 2025**

Related U.S. Application Data

(60) Continuation of application No. 17/885,665, filed on Aug. 11, 2022, which is a division of application No. 17/534,001, filed on Nov. 23, 2021, now Pat. No. 11,439,386, which is a continuation-in-part of application No. 16/537,088, filed on Aug. 9, 2019, now Pat. No. 11,213,290.





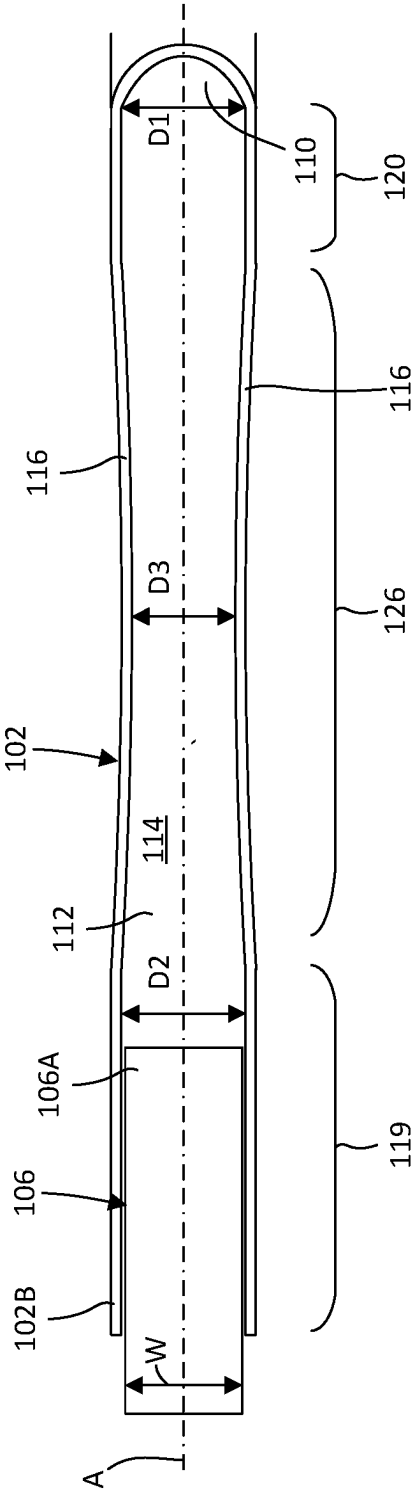


FIG. 3

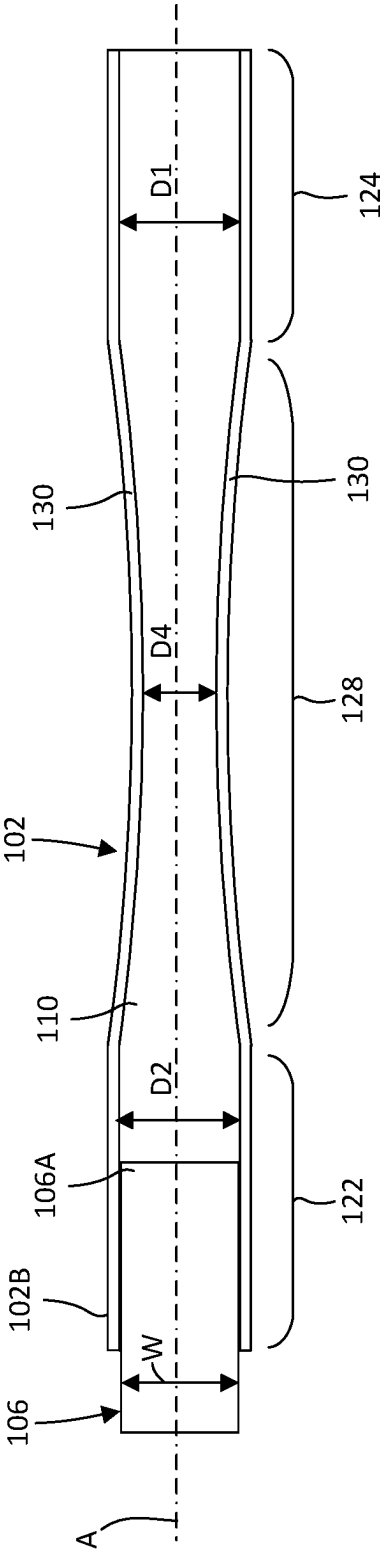


FIG. 4

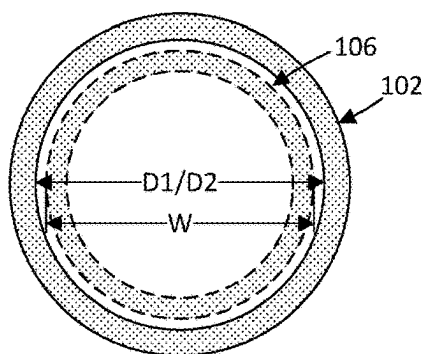


FIG. 5

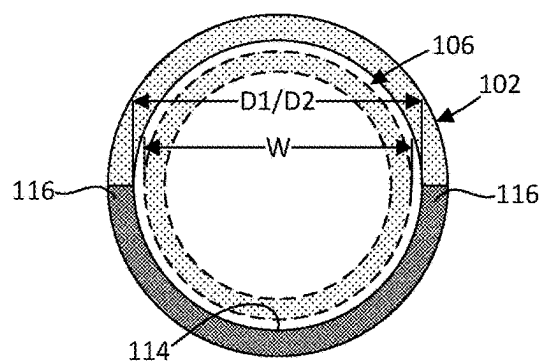


FIG. 6A

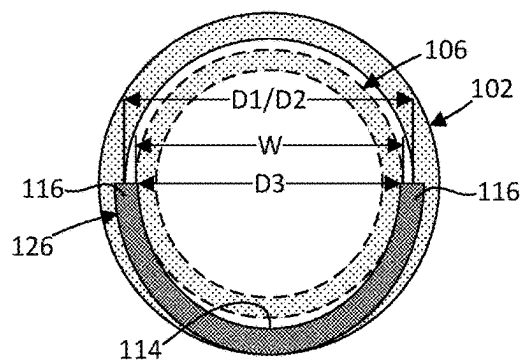


FIG. 6B

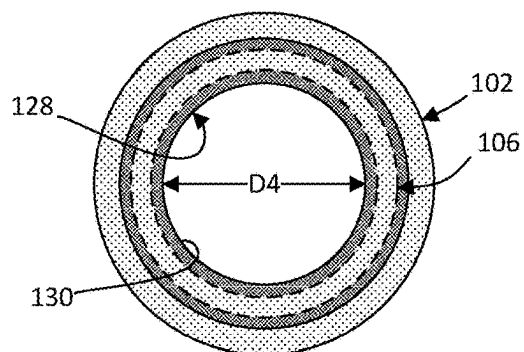


FIG. 7

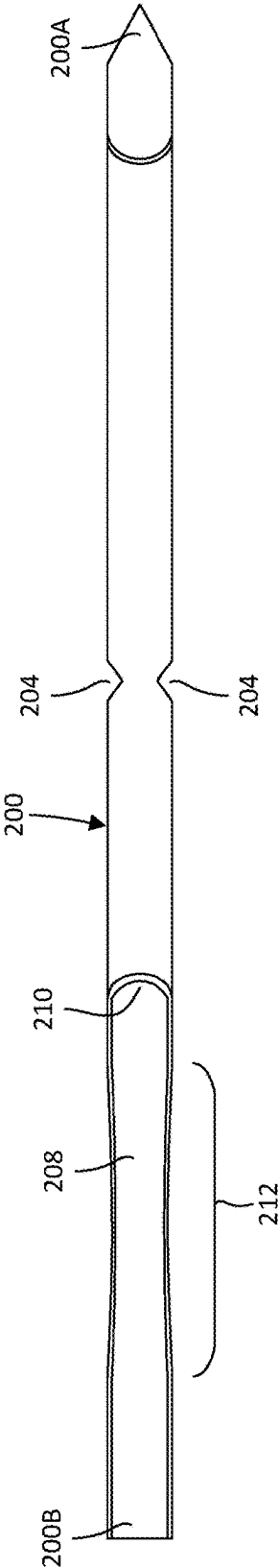


FIG. 8

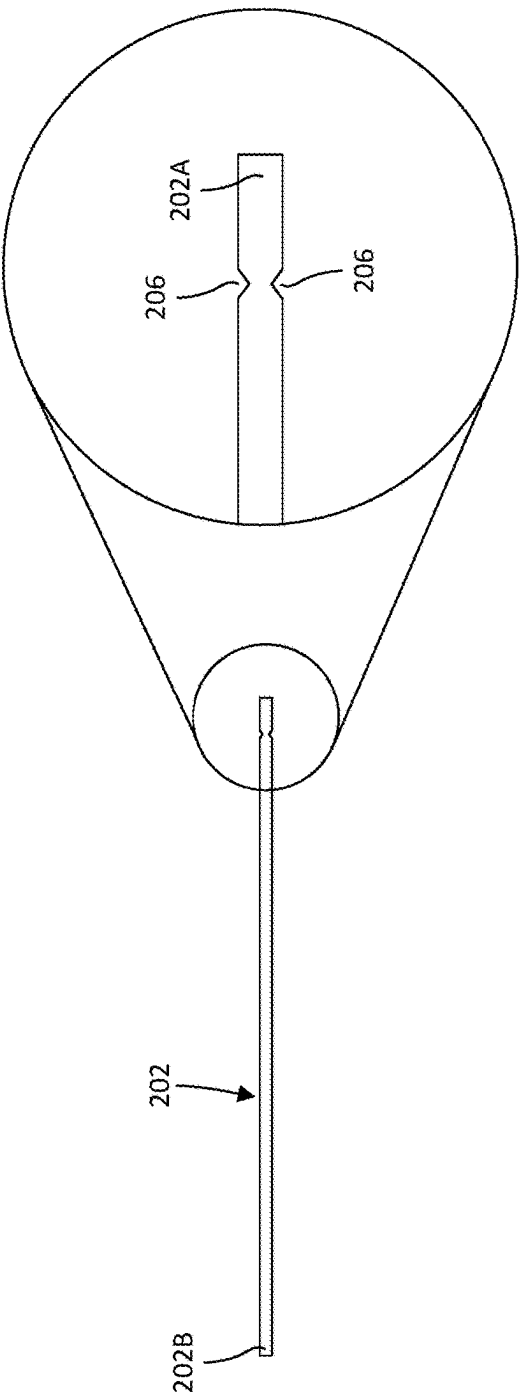


FIG. 9

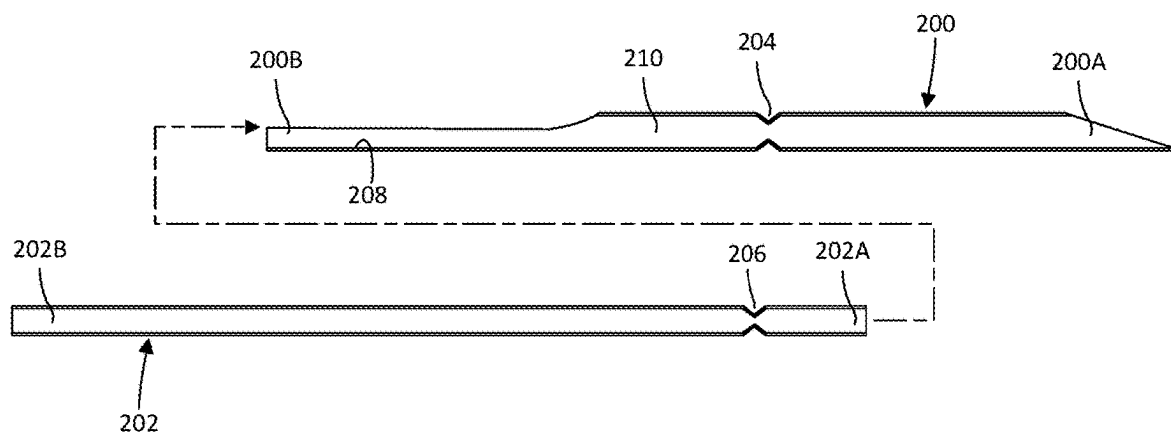


FIG. 10

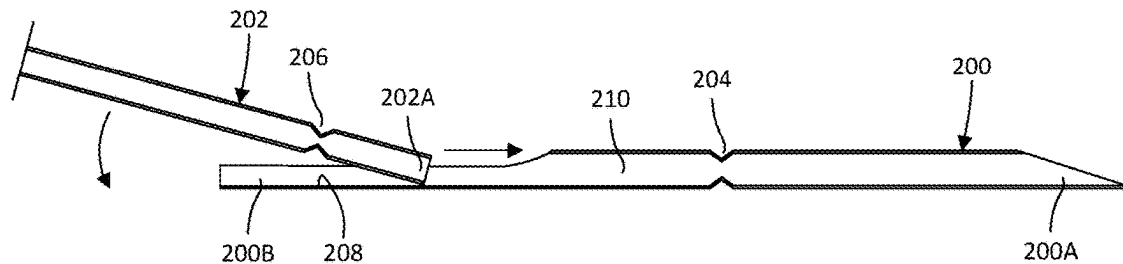


FIG. 11

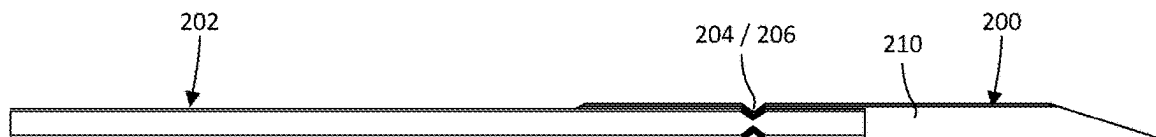


FIG. 12

INTERLOCKING TWO-PART STRUCTURES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation-in-part of application Ser. No. 16/537,088, filed Aug. 9, 2019, which is a continuation-in-part of application Ser. No. 16/151,591, filed Oct. 4, 2018, now U.S. Pat. No. 10,792,036, which claims priority to U.S. Provisional Patent Application No. 62/586,690, filed on Nov. 15, 2017, each of the foregoing applications entitled METHODS AND SYSTEMS FOR DOUBLE LOOP STITCHING; the entire contents of all of the foregoing applications being incorporated herein by reference.

FIELD OF THE INVENTION

[0002] This invention generally relates to a method and apparatus for stitching. More specifically, the present invention, relates to an interlocking two-part sewing needle including a first needle portion having an integrated needle receiver for guiding the insertion of a second needle portion into engagement with the first needle portion and a needle portion interlock for removably connecting the first needle portion and the second needle portion together.

BACKGROUND OF THE INVENTION

[0003] Ligament replacement or repair is very common amongst athletes and active individuals. In a typical ligament replacement surgery, the torn ligament is replaced by soft tissue graft, which may be taken from the patient's own body or from a donor. During the surgery, the surgeon uses a needle and heavy-duty, non-degradable suture to reinforce the ends of the graft. The reinforced graft is then implanted into the knee and affixed so that it serves the same anatomical function that the torn ligament served prior to the injury.

[0004] The conventional graft preparation process discussed above may be improved through use of the two-part needle and methods that are detailed in U.S. Pat. No. 10,792,036 and application Ser. No. 16/537,088, the contents of which are hereby incorporated by reference in their entirety. That two-part needle and the related methods may be further improved by a two-part needle and method of use that provides for a more secure connection between the two needle portions.

Notes on Construction

[0005] The use of the terms “a”, “an”, “the” and similar terms in the context of describing the invention are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising”, “having”, “including” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. The terms “substantially”, “generally” and other words of degree are relative modifiers intended to indicate permissible variation from the characteristic so modified. The use of such terms in describing a physical or functional characteristic of the invention is not intended to limit such characteristic to the absolute value which the term modifies, but rather to provide an approximation of the value of such physical or functional characteristic.

[0006] Terms concerning attachments, coupling and the like, such as “attached”, “connected” and “interconnected”,

refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both moveable and rigid attachments or relationships, unless specified herein or clearly indicated by context. The term “operatively connected” is such an attachment, coupling or connection that allows the pertinent structures to operate as intended by virtue of that relationship.

[0007] The use of any and all examples or exemplary language (e.g., “such as” and “preferably”) herein is intended merely to better illuminate the invention and the preferred embodiments thereof, and not to place a limitation on the scope of the invention. Nothing in the specification should be construed as indicating any element as essential to the practice of the invention unless so stated with specificity.

[0008] The apparatus and method disclosed herein may be used for traditional sewing processes, such as joining fabric or textiles in the production of clothing, etc., in surgical procedures, and other similar processes. As such, in the description that follows, the term “thread” is used broadly and interchangeably to refer to textile-type strands used in traditional sewing processes, to refer to medical sutures or other similar materials for use in performing surgery, and the like. Similarly, the term “sewing” is used broadly to refer to the traditional sewing type processes or surgical procedures that use needle and thread. Lastly, the term “sewing material” is used broadly to refer to textile or other types of materials used in traditional sewing type processes or to graft and other similar materials used in surgical procedures, such as ligaments, tendons, or other soft tissue.

[0009] Additionally, the apparatus and method disclosed herein may be used in removably connecting together elongate structures. Unless specifically noted otherwise, the term “elongate structure” should be interpreted broadly to refer to any type of tube, cylinder, cannula, conduit, chute, duct, etc. that can be crimped or that can otherwise be provided with one or more crimped sections. These elongate structures may be used inside of the medical field and also outside of the medical field. As one example of a use outside of the medical field, tube-shaped jump rope handles may be removably connected together using the apparatus and method of the present invention.

SUMMARY OF THE INVENTION

[0010] The above and other needs are met by an interlocking two-part sewing needle that includes a first needle portion having a needle tip and a second needle portion configured to be inserted into the first needle portion for removably connecting the first needle portion and the second needle portion together. A needle receiver is formed in the first needle portion and is configured to receive the second needle portion when the second needle portion is inserted into the first needle portion. The needle receiver includes one or more non-resistance sections. Each non-resistance section is configured to exert only a friction force on the second needle portion when the second needle portion is inserted into the needle receiver. Additionally, the needle receiver includes one or more resistance sections. Each resistance section is configured to exert a compressive force on the second needle portion when the second needle portion is inserted into the needle receiver.

[0011] In certain embodiments, the needle receiver includes a channel that, when the second needle portion is inserted into the first needle portion, partially surrounds the

second needle portion. The channel is formed by a bottom that is located between a pair of shoulders that are separated from one another by a first space. In some cases, one or more resistance sections is located in the channel. Further, in some cases, a non-resistance sections is also located in the channel.

[0012] In certain cases, the resistance sections include a channel crimp that is formed by a portion of the shoulders of the channel that are separated from one another by a second space that is less than the first space. The second space is sized such that the portion of the shoulders at the channel crimp exert the compressive force on the second needle when the first needle portion and the second needle portion are connected together. In some embodiments, the non-resistance sections may include a portion of the shoulders located at the channel but not located at the channel crimp that exerts only a friction force on the second needle portion when the first needle portion and the second needle portion are connected together.

[0013] In certain embodiments, the needle receiver includes a bore that is formed and extends longitudinally within the first needle portion. In such cases, the first needle portion and the second needle portion are connected together by sliding the second needle into the bore such that the bore substantially surrounds at least a portion of the second needle portion. In some cases, a resistance section is located in the bore. In some cases, a non-resistance section is also located in the bore. In some embodiments, the first resistance section is sized and configured to arrest sliding movement of the second needle portion with respect to the first needle portion.

[0014] Certain embodiments of the invention include a stop crimp that is formed in the first needle portion that reduces a widest internal wall-to-wall size of the bore taken at a cross section. As a result, when the first needle portion and the second needle portion are connected together, the second needle portion contacts the stop crimp within the bore and the stop crimp exerts the compressive force on the second needle portion. Certain embodiments of the invention include a stop crimp receiver, which is provided in the second needle portion. In those cases, when the first needle portion and the second needle portion are connected together, the stop crimp is configured to snap into the stop crimp receiver for removably connect the first needle portion and the second needle portion together.

[0015] In certain embodiments of the invention, the resistance section includes a first resistance section that is configured to exert a first compressive force on the second needle portion and a second and separate resistance section that is configured to exert a second compressive force on a different and longitudinally spaced apart portion of the second needle portion. In certain embodiments of the invention, at least one of the first needle portion and the second needle portion is formed from a hollow tube. In certain cases, one end of a thread is connected to a trailing end of the first needle portion and an opposing end of the thread is connected to a trailing end of the second needle portion. In certain embodiments of the invention, the compressive force is greater than the friction force exerted on the second needle portion.

[0016] Also disclosed herein is an interlocking two-part sewing needle that includes a first needle portion formed from a hollow tube with an internal bore that extends from a leading end of the first needle portion to a trailing end of

the first needle portion to provide open ends. The sewing needle also includes a channel that is located at the trailing end of the first needle portion. The channel includes a bottom located between a pair of shoulders that are separated from one another by a first space. A needle tip is provided at the leading end of the first needle portion. The two-part sewing needle also includes a second needle portion that is configured to be removably connected together with the first needle portion by inserting a leading end of the second needle portion into the internal bore via the trailing end of the first needle portion and then sliding the second needle portion at least partially through the internal bore. A channel crimp is provided in a portion of the channel. A portion of the shoulders located at the channel crimp are separated by a second space that is smaller than the first space. As a result, the shoulders contact the second needle portion at the channel crimp and align the second needle portion with the internal bore. The two-part needle also includes a stop crimp that constricts the bore and exerts a compressive force on the second needle portion for resisting movement of the second needle portion through the bore beyond the stop crimp.

[0017] Also disclosed herein is a method of using a needle. The method requires the step of providing a two-part sewing needle that includes a first needle portion having a needle tip, a second needle portion configured to be inserted into the first needle portion for removably connecting the first needle portion and the second needle portion together, and a needle receiver formed in the first needle portion that is configured to receive the second needle portion when the second needle portion is inserted into the first needle portion. The needle receiver includes one or more non-resistance sections that are each configured to exert only a friction force on the second needle portion when the second needle portion is inserted into the needle receiver. Additionally the needle receiver also includes one or more resistance sections that are each configured to exert a compressive force on the second needle portion when the second needle portion is inserted into the needle receiver. The method further requires the step of removably connecting the first needle portion and the second needle portion together by inserting the second needle portion into the needle receiver such that a friction force is exerted onto the second needle portion by one of the one or more non-resistance sections. Finally, the method requires the step of, with one of the one or more resistance sections, exerting a first compressive force on the second needle portion that is greater than the friction force exerted on the second needle portion.

[0018] In certain embodiments, the method also includes the step of, with another one of the one or more resistance sections, exerting a second compressive force on a different and longitudinally spaced apart portion of the second needle portion that is greater than the first compressive force and greater than the friction force.

[0019] In certain embodiments, the first compressive force is provided by a channel crimp formed in a portion of the channel, where a portion of the shoulders located at the channel crimp is separated by a second space that is smaller than the first space such that the shoulders contact the second needle portion at the channel crimp and exert the first compressive force on the second needle portion and aligns the second needle portion with the bore. Additionally, the second compressive force is provided by a stop crimp that constricts the bore and exerts the second compressive force

on the second needle portion and resists movement of the second needle portion through the bore beyond the stop crimp.

[0020] Lastly, the present disclosure also provides a method for removably connecting together elongate structures that includes the step of providing a first elongate structure having a longitudinal axis, a leading end, a trailing end, and an elongate receiver having an opening formed at the trailing end of the elongate structure and extending along the longitudinal axis. Additionally, the method includes the step of providing a second elongate structure having a longitudinal axis, a leading end, and a trailing end. Next, the leading end of the second elongate structure is inserted into the opening of the elongate receiver of the first elongate structure such that the longitudinal axes of the first and second elongate structures are parallel. Then, the method includes the step of sliding the second elongate structure into the elongate receiver. A non-resistance section is provided in the elongate receiver of the first elongate structure. With the non-resistance section, only a friction force is exerted on the second elongate structure as the second elongate structure is sliding into the elongate receiver. The method also includes the step of providing a resistance section in the elongate receiver of the first elongate structure. With the resistance section, a compression force is exerted on the second elongate structure as the second elongate structure is inserted into the elongate receiver.

[0021] In certain embodiments, the first elongate structure includes a hollow tube having an internal bore formed by an outer wall providing opposing open ends. Additionally, the resistance section includes a crimp formed in the outer wall of the hollow tube that reduces a widest internal wall-to-wall spacing of the internal bore located at the crimp and taken at a cross section. Lastly, the second elongate structure has a width that is greater than the widest internal wall-to-wall spacing. The method further includes the step of exerting a compressive force onto the second elongate structure by contacting the second elongate structure with an internal wall surface located at the crimp. In certain cases, the second elongate structure includes an outer wall and a crimp receiver formed in the outer wall and the method further includes the step of positively engaging a portion of the outer wall located at the crimp into the crimp receiver. In certain embodiments, the first and second elongate structures each include a hollow tube, each having an internal bore formed by an outer wall providing opposing open ends.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The presently preferred embodiments of the invention are illustrated in the accompanying drawings, in which like reference numerals represent like parts throughout, and in which:

[0023] FIG. 1 is a plan view depicting a crimped two-part needle according to an embodiment of the present invention, where first and second needle portions, which are connected together by a suture/thread, are shown in a separated configuration;

[0024] FIG. 2 is a plan view depicting the needle of FIG. 1, where the first needle portion and the second needle portion are shown in a connected and locked together configuration;

[0025] FIG. 3 is a plan view depicting a first resistance section of the first needle portion of FIG. 1 that is enclosed by a box indicated by "FIG. 3";

[0026] FIG. 4 is a cutaway view depicting the internal structure of a second resistance section of the first needle portion of FIG. 1 that is enclosed by a box indicated by "FIG. 4";

[0027] FIG. 5 is a cross sectional view showing a cross section of the first needle portion and the second needle portion of FIG. 3;

[0028] FIG. 6A is a cross sectional view showing a cross section of the first needle portion and the second needle portion of FIG. 3 and also a cross section of a channel shown in FIG. 3;

[0029] FIG. 6B is a cross sectional view showing a cross section of the first needle portion and the second needle portion of FIG. 3 and also a cross section of a channel crimp shown in FIG. 3;

[0030] FIG. 7 is a cross sectional view showing a cross section of the first needle portion and the second needle portion of FIG. 4 and also a cross section of a stop crimp shown in FIG. 4;

[0031] FIG. 8 is a plan view depicting a crimped first needle portion according to an alternative embodiment of the present invention;

[0032] FIG. 9 is detail view depicting a second needle portion having a crimp receiver according to an alternative embodiment of the present invention;

[0033] FIG. 10 depicts the first needle portion and the second needle portion of FIGS. 8 and 9 in a separated configuration;

[0034] FIG. 11 depicts the first needle portion and the second needle portion of FIG. 10 in the process of being connected together; and

[0035] FIG. 12 depicts the first needle portion and the second needle portion of FIG. 10 in a connected and locked together configuration.

DESCRIPTION OF THE PREFERRED EMBODIMENTS OF THE INVENTION

[0036] This description of the preferred embodiments of the invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description of this invention. The drawings are not necessarily to scale, and certain features of the invention may be shown exaggerated in scale or in somewhat schematic form in the interest of clarity and conciseness.

[0037] With initial reference to FIGS. 1, 2, and 5 there is provided an interlocking two-part sewing needle 100 according to an embodiment of the present invention. The needle 100 includes a first needle portion 102 having a leading end 102A, a trailing end 102B, and a needle tip 104 located at the leading end; and a second needle portion 106 having a leading end 106A and a trailing end 106B. In certain embodiments, the first needle portion 102 is a hollow tube. In certain other embodiments, including the illustrated embodiment, both the first needle portion and the second needle portion 102, 106 are formed from hollow tubes. The second needle portion 106 is configured to be inserted into the first needle portion 102 in order to removably connect the first needle portion and the second needle portion together during the sewing process (e.g., prior or during a ligament replacement surgery). The needle 100 may also include a thread 118 (e.g., a suture), wherein one end of the thread is connected to the trailing end 102B of the first needle portion 102 and an opposing end of the thread is

connected to a trailing end **106B** of the second needle portion **106**. This combined needle **100** and thread **118** may be used according to the methods described in U.S. Pat. No. 10,792,036 and application Ser. No. 16/537,088.

[0038] A needle receiver **108** is formed in the first needle portion **102** and is configured to receive the second needle portion **106** when the first needle portion and the second needle portion are connected together. The second needle portion **106** may also be selectively removed from the needle receiver **108** when the first needle portion and the second needle portion **102**, **106** are separated from one another. The second needle portion **106** may be inserted into and removed from any portion of the first needle portion **102**. For example, in certain embodiments, the needle receiver **108** may include a slot or indentation (not shown) formed in the outside of the side wall of the first needle portion **102**. The slot is preferably sized and configured to allow the second needle portion **106** to be inserted and, preferably, removably attached to the first needle portion **102**, such as by clips or other similar retention means integrated into the slot.

[0039] However, in preferred embodiments, the second needle portion **106** is inserted longitudinally into an opening formed in the first needle portion **102**. As mentioned above, the first needle portion and the second needle portion **102**, **106** are formed from hollow tubes, such that openings are located at each end of both needle portions **102**, **106**. These tubes each include a bore **110** that extends longitudinally through the tube and that forms part of the needle receiver **108**. The bore has an opening having a first space **D1** (i.e., the internal diameter or the widest internal wall-to-wall spacing separating taken at a cross section).

[0040] In other embodiments, the bore **110** extends only partially through the needle portions **102**, **106**, such that only one end is provided with an opening. In preferred embodiments, the bore **110** of the first needle portion **102** is sized and configured such that the second needle portion **106** may be partially or fully inserted into the bore. As more fully described below, when the second needle portion **106** is inserted into the first needle portion **102**, the bore **110** of the first needle portion “substantially” surrounds the inserted portion of the second needle portion.

[0041] Next, the portion of the needle **100** shown in FIG. 1 that is enclosed by a box labeled “FIG. 3” is shown and enlarged in FIG. 3. In FIG. 3, the leading end **106A** of the second needle portion **106** is engaged with a portion of the trailing end **102B** of the first needle portion **102**. In particular, in preferred embodiments, the needle receiver **108** also includes a channel **112** that is located at the trailing end **102B** of the first needle portion and is configured to receive the second needle portion **106**, as shown in FIG. 3. With continued reference to FIG. 3 and with further reference to FIG. 6A, the channel **112** is formed by a bottom **114** located between a pair of shoulders **116** that are preferably separated from one another by a second space **D2** (i.e., the widest internal wall-to-wall spacing separating taken at a cross section), which may be equal to space **D1**. However, depending on the size of the shoulders **116**, the spacing between the shoulders may be more or less than space **D1**. The channel **112** is sized and configured to receive and to “partially” surround the second needle portion **106** when the second needle portion is connected together with the first needle portion **102**.

[0042] When “partially” surrounded by just the shoulders **116**, the second needle portion **106** is not secured within the channel **112**. Instead, the second needle portion **106** may be easily inserted and removed (i.e., lifted) from the channel **112** without resistance. In other words, there is sufficient space between the shoulders **116** (i.e. space **D2**) to enable the second needle portion **106** to pass through easily and without resistance. By contrast, when a portion of the second needle portion **106** is inserted into the bore **110** and that inserted portion is “substantially” surrounded by the bore, the inserted portion of the second needle portion is secured within the bore and cannot be removed (i.e., lifted out of the bore) without resistance. In certain cases, the bore **110** fully surrounds the inserted portion of the second needle portion **106**. In other cases, there may be a small slot or opening, similar to the space **D2** between the shoulders **116**. However, that slot or opening provides an insufficient amount of space to allow the inserted portion of the second needle portion **106** to pass through easily or without resistance.

[0043] The channel **112** may be formed by removing a portion of the wall that forms the bore **110**. Alternatively, the channel **112** may be attached to the end of the bore **110**. In either case, the channel **112** is preferably connected continuously with the bore **110** such that the second needle portion **106** can move easily between the bore and channel as it is inserted into and removed from the first needle portion **102**.

[0044] As the first needle portion and the second needle portion **102**, **106** are engaged with one another (i.e., as the second needle portion is inserted into the needle receiver **108**), they are in sliding contact with one another. As a result of this sliding contact, the needle portions **102**, **106** each exert a friction force on the other needle portion. Thus, in preferred embodiments, the needle receiver **108** includes one or more non-resistance sections, where only a friction force is exerted on the second needle portion **106** when the second needle portion interacts with (e.g. is inserted into) the needle receiver of the first needle portion **102**. There may be multiple non-resistance sections in various locations of the needle receiver **108**, including in the bore **110** and in the channel **112**. In FIG. 3, two separate non-resistance sections (the areas generally denoted with reference numbers **119** and **120**) are provided, where the second needle portion **106** experiences only a friction force (if contact is made between the needle portions). Now, with reference to FIG. 4, a cutaway view is provided that shows the portion of the needle **100** in FIG. 1 that is surrounded by a box and that is labeled “FIG. 4.” In FIG. 4, two additional non-resistance sections (the areas generally denoted with reference numbers **122** and **124**) are provided, where the second needle portion **106** experiences only a friction force (again, if contact is made between the needle portions).

[0045] Now, with reference to both FIGS. 3 and 4, the needle **100** also preferably includes one or more resistance sections, where a force that is greater than the above-described friction force is exerted on the second needle portion **106**. These one or more resistance sections function as a “needle interlock”. As discussed in greater detail below, among other things, the needle interlock, via these resistance sections, assists in guiding the second needle portion **106** as it is joined together with the first needle portion **102**. Additionally, the needle interlock provides tactile feedback to help the user to know when the connection between the

needle portions **102**, **106** is complete. Lastly, the needle interlock assists in maintaining the connection between the needle portions **102**, **106**.

[0046] Unlike the friction force experienced by the second needle portion **106** at the non-resistance sections described above, in preferred embodiments, a compressive force is exerted on the second needle portion by each of the resistance sections. For example, with reference to FIG. 3 and FIG. 6B, a first resistance section is located between non-resistance sections **119** and **120** (i.e., the area generally denoted with reference number **126**). In this particular embodiment, the first resistance section is a channel crimp **126** that is formed by a portion of the shoulders **116** of the channel **112**. The portion of the shoulders **116** that form the channel crimp **126** are separated from one another by a third space **D3** (i.e., the widest internal wall-to-wall spacing separating the shoulders taken at a cross section) that is less than the first space **D1**. Additionally, the third space **D3** is smaller than the widest external width **W** (i.e., the outer diameter) of the second needle portion **106**. The portions of the shoulders **116** that form the channel crimp **126** are configured to flex outwards in order to accommodate the second needle portion **106** (i.e., to allow the second needle portion to be inserted through the channel crimp). As a result, when the second needle portion **106** is inserted into the channel crimp **126**, the shoulders **116** flex outwards and the second needle portion is slightly compressed within the channel crimp.

[0047] The channel crimp **126** may be formed by slightly crimping the shoulders **116** of the channel **112** together in order to decrease the distance between them. Other manufacturing methods for providing a section of the channel have a smaller third space **D3** than first space **D1** may also be used to create the channel crimp **126**. The channel crimp **126** may be formed as a narrow crimp in order to compress only a small section of the second needle portion **106**. In other cases, as illustrated, a wider and more gradual crimp may be provided in order to compress a wider section of the second needle portion **106**.

[0048] When in use, the channel crimp **126** initially assists in holding the first needle portion and the second needle portion **102**, **106** together as they are being connected. As the leading end **106A** and the second needle portion **106** is pushed further into the channel **112**, the channel crimp **126** also assists in aligning the second needle portion with the bore **110**. More particularly, the channel crimp **126** preferably automatically locates the longitudinal midline of the second needle portion **106** at a center position between the shoulders (i.e., on longitudinal axis **A**). This enables the second needle portion **106** to pass much more quickly and easily into the bore **110**. Since the needle portions **102**, **106** are connected and then disconnected from one another multiple times during the course of a surgical procedure, this time reduction can result in significant time and cost savings compared to traditional surgical methods.

[0049] Similarly, in preferred embodiments, a second resistance section (the area generally denoted with reference number **128**) is located between non-resistance sections **122** and **124**. Preferably, the second resistance section is longitudinally spaced apart from the first resistance section. In particular, with continued reference to FIG. 4 and with further reference to FIG. 7, a stop crimp **128** is formed by walls **130** that form the bore **110** and that are separated from one another by a fourth space **D4** (i.e., the widest internal

wall-to-wall spacing separating the walls). Unlike the shoulders **116**, which only partially surrounds the second needle portion **106**, the walls **130** that form the bore **110** substantially surround the second needle portion. The stop crimp **128** is created and functions similarly to the channel crimp **126**. In particular, the stop crimp **128** may be formed by crimping the walls **130** of the bore **110** together in order to decrease the distance between them. The stop crimp **128** may be formed as a narrow crimp in order to compress only a small section of the second needle portion **106**. In other cases, as illustrated, a wider and more gradual crimp may be provided in order to compress a wider section of the second needle portion **106**.

[0050] Like the third space **D3**, the fourth space **D4** is also smaller than the width **W** of the second needle portion **106**. The portions of the walls **130** that form the stop crimp **128** are also configured to flex outwards in order to accommodate a portion the second needle portion **106**. However, the stop crimp **128** is configured to exert a greater compressive force on the second needle portion **106** than the channel crimp **126**. The stop crimp **128** is configured to constrict the bore **110** and to exert a compressive force on the second needle portion **106** that is different and preferably greater than the compressive force provided by the channel crimp **126**, so as to limit or, more preferably, arrest the movement of the second needle portion through the bore and beyond the stop crimp. As a result, in preferred embodiments, the second needle portion **106** may be partially inserted into the stop crimp **126** before coming to a stop as a result of the compression from the stop crimp. This provides a tactile “click” sensation to the user when the second needle portion **106** comes to a stop within the stop crimp **128**, which informs the user that the first needle portion **102** and the second needle portion are connected together.

[0051] In the descriptions above, the resistance sections **126**, **128** are each located between a pair of non-resistance sections **119** and **120** and also **122** and **124**, respectively. However, in other embodiments, a pair of resistance sections may be located adjacent one another without a non-resistance section between them. For example, a set of crimps may be provided in the bore **110** that progressively reduce the space within the bore.

[0052] Now, with reference to FIGS. 8 and 9, a first needle portion **200** and a second needle portion **202** according to an alternative embodiment of the present invention are shown. First needle portion **200** is structurally and functionally similar to needle portion **102**, but includes a shortened stop crimp **204** that is more abrupt or angular than stop crimp **128**, which is longer and more gradual in shape. Second needle portion **202** is also structurally and functionally similar to second needle portion **106**, but includes a stop crimp receiver **206** near the leading end **202A** of the second needle portion. The stop receiver **206** is sized and configured to automatically engage with the stop crimp **204** when the second needle portion **202** is placed onto the channel **208** and then inserted into the bore **210** of the first needle portion **200**. This process is illustrated in FIGS. 10-12 and is detailed below.

[0053] The first needle portion and the second needle portion **200**, **202** are initially shown separated from one another (FIG. 10). To connect the needle portions **200**, **202** together, the leading end **202A** of the second needle portion **202** is inserted into the trailing end **200B** of the first needle portion **200** (FIG. 11). The second needle portion **202** then

slides forward towards the leading end **200A** of the first needle portion, including through the channel **208** and at least a portion of the bore **210**. Preferably, this sliding motion is guided by a channel crimp **212** (FIG. 8), as described previously. The second needle portion **202** continues sliding free through the bore **210**, guided by the channel crimp, until it reaches the shortened stop crimp **204**. At that point, the leading end **202A** of the second needle portion **202** contacts the stop crimp **204**, which provides resistance to continued forward movement. Pushing the second needle portion **202** through the bore **210** causes the walls of the first needle portion **200** to flex outwards, the walls of the second needle portion **202** to flex inwards, or both, which allows the leading end **202A** of the second needle portion to continue sliding beyond the stop crimp **204**. The walls of the needle portions **200**, **202** remain in flexed state until the stop crimp **204** snaps into engagement with the stop crimp receiver **206** (FIG. 12). This snapping action occurs when the walls of the needle portions **200**, **202** return to an unflexed state. Advantageously, the snapping action also provides tactile feedback to the user that the needle portions **200**, **202** are removably engaged with one another.

[0054] More generally, the present invention provides a method for removably connecting any type of elongate structures together. The method includes the step of first providing a first elongate structure having a longitudinal axis, a leading end, a trailing end, and an elongate receiver having an opening formed at the trailing end of the elongate structure and extending along the longitudinal axis. Next, a second elongate structure having a longitudinal axis, a leading end, and a trailing end is provided. The leading end of the second elongate structure is then inserted into the opening of the receiver of the first elongate structure such that the longitudinal axes of the first and second elongate structures are parallel with one another. Then, the second elongate structure is inserted (e.g., slides) into the elongate receiver of the first elongate structure. The method also requires providing a non-resistance section in the receiver of the first elongate structure, where only a friction force is exerted on the second elongate structure as the second elongate structure is inserted into the needle receiver. Additionally, the method requires providing a resistance section in the receiver of the first elongate structure, where a compression force is exerted on the second elongate structure as the second elongate structure is inserted into the needle receiver.

[0055] In certain cases, the first elongate structure is a hollow tube having an internal bore formed by an outer wall. In those cases, the resistance section may include a crimp that is formed in the outer wall of the hollow tube. The crimp reduces the widest internal wall-to-wall spacing of the internal bore located at the crimp taken at a cross section of the internal bore of first elongate structure. Notably, the second elongate structure has a width that is greater than the widest internal wall-to-wall spacing taken at a cross section. As a result, as the second elongate structure is inserted into the receiver of the first elongate structure, a compressive force is exerted onto the second elongate structure by contacting the second elongate structure with an internal wall surface located at the crimp. In certain embodiments, the second elongate structure also includes an outer wall and a crimp receiver formed in the outer wall. As the elongate structures are connected together, the crimp positively

engages the crimp receiver in order to provide a removable connection between them. This also provides tactile feedback to the user when the connection has been made. Lastly, in certain cases, both the first and second elongate structures are formed from hollow tubes.

[0056] Although this description contains many specifics, these should not be construed as limiting the scope of the invention but as merely providing illustrations of some of the presently preferred embodiments thereof, as well as the best mode contemplated by the inventor of carrying out the invention. The invention, as described herein, is susceptible to various modifications and adaptations as would be appreciated by those having ordinary skill in the art to which the invention relates.

What is claimed is:

1. An interlocking two-part sewing needle comprising:
 - a outer needle portion having a needle tip;
 - a inner needle portion configured to be inserted into the outer needle portion for removably connecting the outer needle portion and the inner needle portion together; and
 - a needle receiver formed in the outer needle portion and configured to receive the inner needle portion when the inner needle portion is inserted into the outer needle portion, the needle receiver including:
 - one or more non-resistance sections, each configured to exert only a friction force on the inner needle portion when the inner needle portion is inserted into the needle receiver; and
 - a needle interlock comprising one or more resistance sections, each configured to exert a compressive force on the inner needle portion when the inner needle portion is inserted into the needle receiver.
2. The interlocking two-part sewing needle of claim 1 wherein the needle receiver comprises a channel that, when the inner needle portion is inserted into the outer needle portion, partially surrounds the inner needle portion, the channel formed by a bottom located between a pair of shoulders that are separated from one another by a first space.
3. The interlocking two-part sewing needle of claim 2 wherein one of the one or more resistance sections is located in the channel.
4. The interlocking two-part sewing needle of claim 3 wherein one of the one or more non-resistance sections is located in the channel.
5. The interlocking two-part sewing needle of claim 2 wherein one of the one or more resistance sections of the needle interlock comprises a channel crimp formed by a portion of the shoulders of the channel that are separated from one another by a second space that is less than the first space, wherein the second space is sized such that the portion of the shoulders at the channel crimp exert the compressive force on the inner needle portion when the outer needle portion and the inner needle portion are connected together.
6. The interlocking two-part sewing needle of claim 5 wherein one of the one or more non-resistance sections comprises a portion of the shoulders located at the channel but not located at the channel crimp that exerts only a friction force on the inner needle portion when the outer needle portion and the inner needle portion are connected together.

7. The interlocking two-part sewing needle of claim 1 wherein the needle receiver comprises a bore formed and extending longitudinally within the outer needle portion such that the outer needle portion and the inner needle portion are connected together by sliding the inner needle portion into the bore such that the bore substantially surrounds at least a portion of the inner needle portion.

8. The interlocking two-part sewing needle of claim 7 wherein one of the one or more resistance sections is located in the bore.

9. The interlocking two-part sewing needle of claim 8 wherein one of the one or more non-resistance sections is located in the bore.

10. The interlocking two-part sewing needle of claim 8 wherein the one resistance section is sized and configured to arrest sliding movement of the inner needle portion with respect to the outer needle portion.

11. The interlocking two-part sewing needle of claim 7 further comprising a stop crimp formed in the outer needle portion and reducing a widest internal wall-to-wall size of the bore taken at a cross section such that, when the outer needle portion and the inner needle portion are connected together, the inner needle portion contacts the stop crimp within the bore and the stop crimp exerts the compressive force on the inner needle portion.

12. The interlocking two-part sewing needle of claim 11 further comprising a stop crimp receiver provided in the inner needle portion, wherein, when the outer needle portion and the inner needle portion are connected together, the stop crimp is configured to snap into the stop crimp receiver for removably connect the outer needle portion and the inner needle portion together.

13. (canceled)

14. (canceled)

15. (canceled)

16. (canceled)

17. A interlocking two-part sewing needle comprising:

a outer needle portion formed from a hollow tube with an internal bore that extends from a leading end of the outer needle portion to a trailing end of the outer needle portion to provide open ends; a channel located at the trailing end of the outer needle portion and including a bottom located between a pair of shoulders that are separated from one another by a first space; and a needle tip provided at the leading end of the outer needle portion;

a inner needle portion configured to be removably connected together with the outer needle portion by inserting a leading end of the inner needle portion into the internal bore via the trailing end of the outer needle portion and then sliding the inner needle portion at least partially through the internal bore;

a channel crimp provided in a portion of the channel wherein a portion of the shoulders located at the channel crimp are separated by a second space that is smaller than the first space such that the shoulders contact the inner needle portion at the channel crimp and align the inner needle portion with the internal bore; and

a stop crimp that constricts the internal bore and exerts a compressive force on the inner needle portion for resisting movement of the inner needle portion through the internal bore beyond the stop crimp.

18. A method of using a needle comprising the steps of: providing a two-part sewing needle comprised by a outer needle portion having a needle tip; a inner needle portion configured to be inserted into the outer needle portion for removably connecting the outer needle portion and the inner needle portion together; and a needle receiver formed in the outer needle portion and configured to receive the inner needle portion when the inner needle portion is inserted into the outer needle portion, the needle receiver including: one or more non-resistance sections, each configured to exert only a friction force on the inner needle portion when the inner needle portion is inserted into the needle receiver; and a needle interlock having one or more resistance sections, each configured to exert a compressive force on the inner needle portion when the inner needle portion is inserted into the needle receiver;

removably connecting the outer needle portion and the inner needle portion together by inserting the inner needle portion into the needle receiver such that a friction force is exerted onto the inner needle portion by one of the one or more non-resistance sections; and

with one of the one or more resistance sections, exerting a first compressive force on the inner needle portion that is greater than the friction force exerted on the inner needle portion.

19. The method of claim 18 further comprising the step of, with another one of the one or more resistance sections, exerting a second compressive force on a different and longitudinally spaced apart portion of the inner needle portion that is greater than the first compressive force and greater than the friction force.

20. The method of claim 19 wherein:

the needle receiver comprises:

a channel that, when the inner needle portion is inserted into the needle receiver, partially surrounds the inner needle portion, the channel formed by a bottom located between a pair of shoulders that are separated from one another by a first space;

a bore formed and extending longitudinally within the outer needle portion such that the outer needle portion and the inner needle portion are connected together by sliding the inner needle portion into the bore such that the bore substantially surrounds at least a portion of the inner needle portion;

the first compressive force is provided by a channel crimp formed in a portion of the channel, wherein a portion of the shoulders located at the channel crimp is separated by a second space that is smaller than the first space such that the shoulders contact the inner needle portion at the channel crimp and exert the first compressive force on the inner needle portion and aligns the inner needle portion with the bore; and

the second compressive force is provided by a stop crimp that constricts the bore and exerts the second compressive force on the inner needle portion and resists movement of the inner needle portion through the bore beyond the stop crimp.

21. A method for removably connecting together elongate structures comprising the steps of:

providing a outer needle portion having a longitudinal axis, a leading end, a trailing end, and an elongate

receiver having an opening formed at the trailing end of the outer needle portion and extending along the longitudinal axis;

providing a inner needle portion having a longitudinal axis, a leading end, and a trailing end;

inserting the leading end of the inner needle portion into the opening of the elongate receiver of the outer needle portion such that the longitudinal axes of the inner and outer needle portions are parallel and then sliding the inner needle portion into the elongate receiver;

providing a non-resistance section in the elongate receiver of the outer needle portion;

with the non-resistance section, exerting only a friction force on the inner needle portion as the inner needle portion is sliding into the elongate receiver;

providing a resistance section in the elongate receiver of the outer needle portion; and

with the resistance section, exerting a compression force on the inner needle portion as the inner needle portion is inserted into the elongate receiver.

22. The method of claim **21** wherein:

the outer needle portion comprises a hollow tube having an internal bore formed by an outer wall providing opposing open ends,

the resistance section comprises a crimp formed in the outer wall of the hollow tube that reduces a widest internal wall-to-wall spacing of the internal bore located at the crimp and taken at a cross section,

the inner needle portion has a width that is greater than the widest internal wall-to-wall spacing, and

the method further comprising the step of exerting a compressive force onto the inner needle portion by contacting the inner needle portion with an internal wall surface located at the crimp.

23. The method of claim **22** wherein the inner needle portion comprises an outer wall and a crimp receiver formed in the outer wall, the method further comprising the step of positively engaging a portion of the outer wall located at the crimp into the crimp receiver.

24. The method of claim **21** wherein the inner and outer needle portions each comprise a hollow tube, each having an internal bore formed by an outer wall providing opposing open ends.

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